

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2022 - 2023)

JEE(Main)

UNIT TEST

30-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS & TNAS]

Time : 3 Hours

PAPER-2

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in JEE(Main) 2023Corporate Office : **ALLEN** CAREER INSTITUTE, "SANKALP", CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005

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SECTION-I : (Maximum Marks: 80)

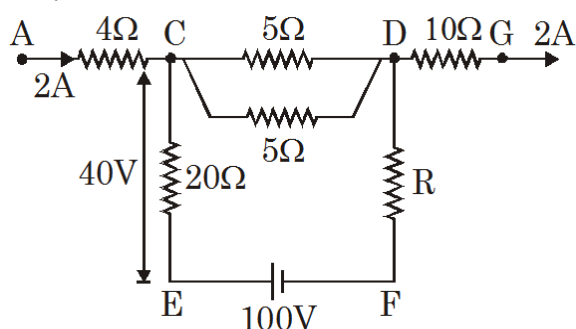
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

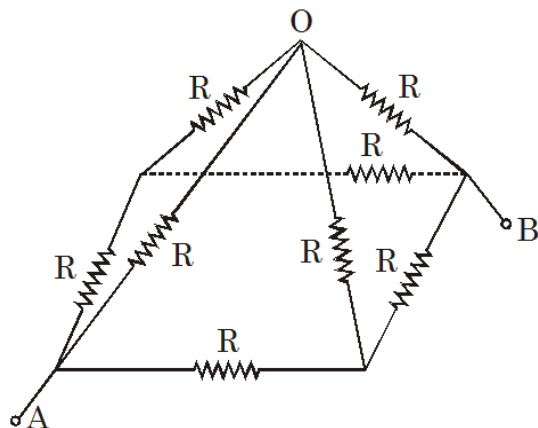
Negative Marks : -1 If wrong option is selected.

1. In the given circuit, find the value of resistance 'R' :



- (A) 20Ω (B) 25Ω
(C) 15Ω (D) 10Ω

2. In the given figure, $R = 3\Omega$. Determine equivalent resistance between points A and B.

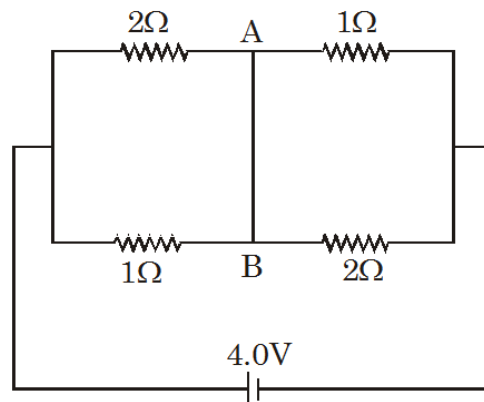


- (A) 3 Ω (B) 2 Ω
(C) 1.5 Ω (D) 6 Ω

3. A piece of conducting wire of resistance R is cut into $2n$ equal parts. Half the parts are connected in series to form a bundle and remaining half in parallel to form another bundle. These bundles are then connected to give the maximum resistance. The resistance of the combination is

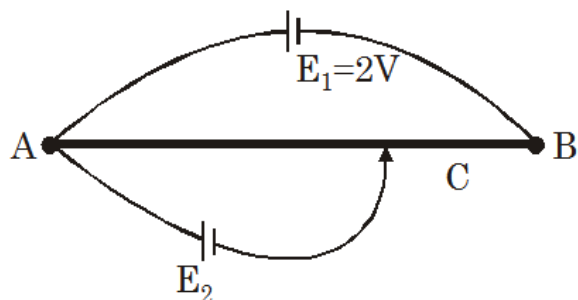
- (A) $\frac{R}{2} \left(1 + \frac{1}{n^2} \right)$
(B) $\frac{R}{2} (1 + n^2)$
(C) $\frac{R}{2(1 + n^2)}$
(D) $R \left(n + \frac{1}{n} \right)$

4. In figure, points A and B are connected by a perfectly conducting wire. Calculate the current through AB



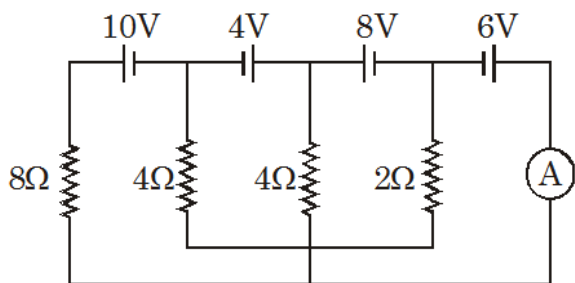
- (A) 2A
(B) 1 A
(C) 1.5 A
(D) 2.5 A

5. AB is a potentiometer wire of length 100 cm. When a cell E_1 is connected across AC, where $AC = 75$ cm, no current flows from E_2 . The internal resistance of the cell E_1 is negligible.



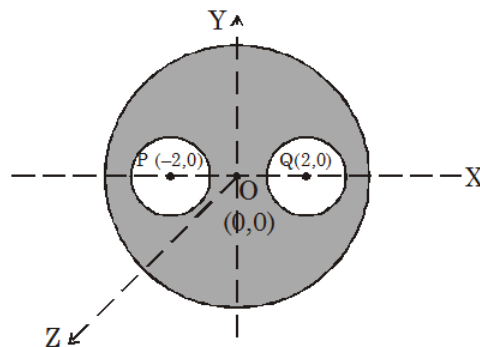
Potential gradient along AB and value of E_2 respectively is.

- (A) 0.01 V cm^{-1} , 1.2 volt
 (B) 0.03 V cm^{-1} , 1.3 volt
 (C) 0.04 V m^{-1} , 1.0 volt
 (D) 0.02 V cm^{-1} , 1.5 volt
6. Find the reading of the ideal ammeter connected in the given circuit. Assume that the cells have negligible internal resistance.

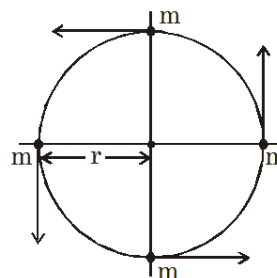


- (A) 2.5 A
 (B) 5 A
 (C) 2 A
 (D) none of these

7. A solid sphere of uniform density & having total mass (M) now two spherical cavity is removed as shown in figure. Radius of sphere is 4 meter. Radius of each spherical cavity is 1 meter. Value of gravitation field at centre of sphere will be :-

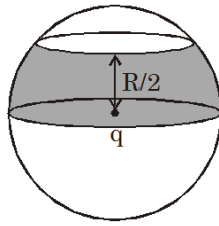


- (A) $\frac{31GM}{1024}$ (B) $\frac{31GM}{256}$
 (C) $\frac{GM}{1024}$ (D) Zero
8. Four similar masses orbiting around a circle of radius 'r'. Due to their mutual gravitation attraction force velocity of each particle will be :-

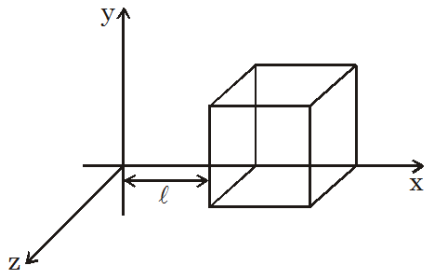


- (A) $\sqrt{\frac{Gm}{r} \left(\frac{1 + 2\sqrt{2}}{4} \right)}$
 (B) $\sqrt{\frac{Gm}{r} (1 + 2\sqrt{2})}$
 (C) $\sqrt{\frac{Gm}{r}}$
 (D) $\sqrt{\frac{Gm}{r} \left(\frac{1 + 2\sqrt{2}}{2} \right)}$

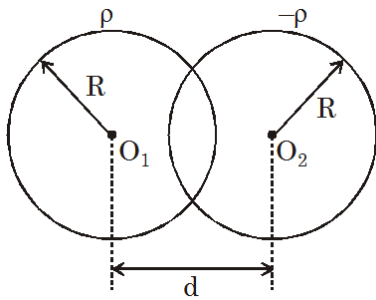
9. Find out flux passing through shaded surface of the sphere when a point charge 'q' is placed at centre of sphere. [Radius of sphere is R]



- (A) $\frac{q}{\epsilon_0}$ (B) $\frac{q}{4\epsilon_0}$ (C) $\frac{q}{2\epsilon_0}$ (D) Zero.
10. A cube of side ' ℓ ' placed as shown in figure $E_x = \beta x^{\frac{1}{2}}$. Flux passing through cube will be:-

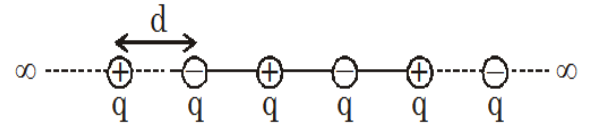


- (A) $2\sqrt{2}\beta\ell^{\frac{5}{2}}$ (B) $(\sqrt{2} - 1)\beta\ell^{\frac{5}{2}}$
 (C) $-\beta\ell^{\frac{5}{2}}$ (D) Zero
11. Two solid spheres having volume charge density ρ and $-\rho$ respectively as shown in figure. Find out the potential difference (ΔV) between centre of both sphere for $d = R$

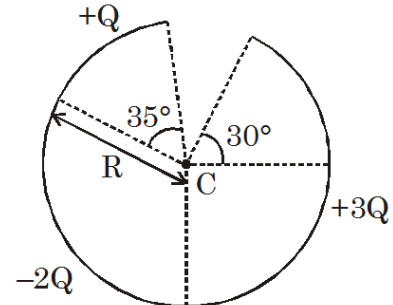


- (A) $\frac{\rho}{3\epsilon_0}R^2$ (B) $\frac{\rho}{\epsilon_0}R^2$ (C) 0 (D) $\frac{2\rho}{\epsilon_0}R^2$

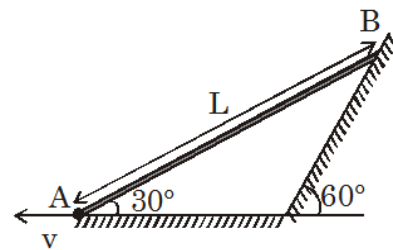
12. There is a infinite chain of charge q & -q. Distance between the two neighbouring charge is equal to 'd'. The interaction energy of one of charge with all other charge is :



- (A) $\frac{-2q^2}{4\pi\epsilon_0 d}$ (B) $\frac{-2q^2 \ln 2}{4\pi\epsilon_0 d}$
 (C) $\frac{2q^2 \ln 2}{4\pi\epsilon_0 d}$ (D) $\frac{q^2}{\pi\epsilon_0 d}$
13. Three arc are joined together as shown in figure having charges Q, -2Q, 3Q respectively. Find out the electric potential at point C.

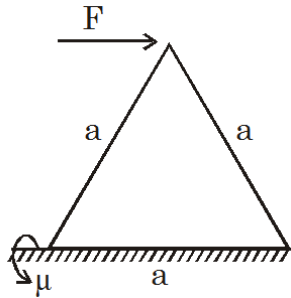


- (A) $\frac{Q}{2\pi\epsilon_0 R}$ (B) $\frac{5Q}{12\pi\epsilon_0 R}$
 (C) $\frac{3Q}{32\pi\epsilon_0 R}$ (D) $\frac{3Q}{8\pi\epsilon_0 R}$
14. Instantaneous speed on end A of rod is 'v'. Angular velocity of rod of length 'L' will be.

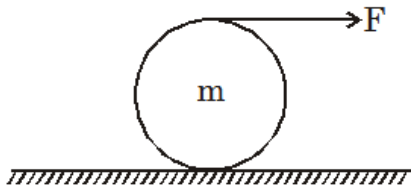


- (A) $\frac{v}{2L}$ (B) $\frac{v\sqrt{3}}{2L}$ (C) $\frac{v}{L}$ (D) $\frac{4v}{2L}$

15. Equilateral prism of mass m is rest on rough horizontal floor. If coefficient of friction is sufficiently high such that prism does not slide before topple. Minimum force required to topple prism is (coefficient of friction = μ)



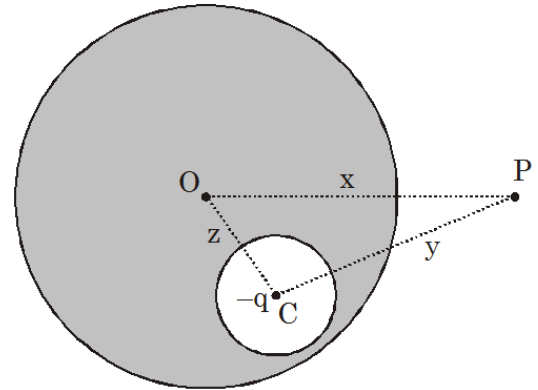
- (A) $\frac{mg}{\sqrt{3}}$ (B) $\frac{\mu mg}{\sqrt{3}}$
 (C) $\frac{mg}{4}$ (D) $\frac{\mu mg}{4}$
16. A force 'F' act tangentially at highest point of a disc of mass 'm' & radius 'R'. If disc is rolling without sliding. Acceleration of centre of disc will be



- (A) $\frac{2F}{3m}$ (B) 0 (C) $\frac{10F}{7m}$ (D) $\frac{4F}{3m}$
17. Time period of a simple pendulum whose length is $\frac{R}{2}$, where R is radius of the earth oscillating near earth's surface will be -

- (A) $2\pi\sqrt{\frac{R}{2g}}$ (B) $2\pi\sqrt{\frac{R}{g}}$
 (C) $2\pi\sqrt{\frac{R}{3g}}$ (D) $2\pi\sqrt{\frac{3R}{g}}$

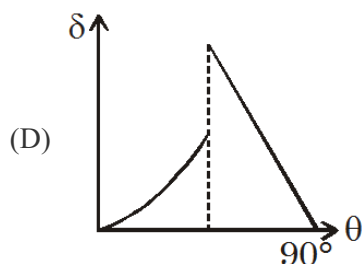
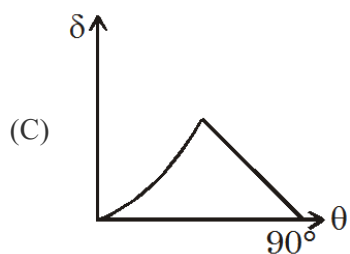
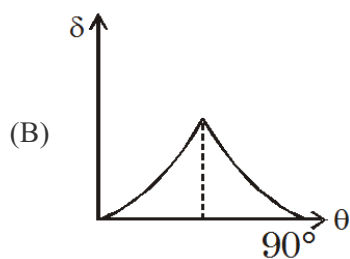
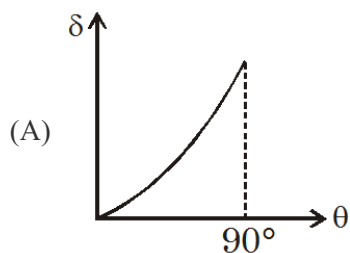
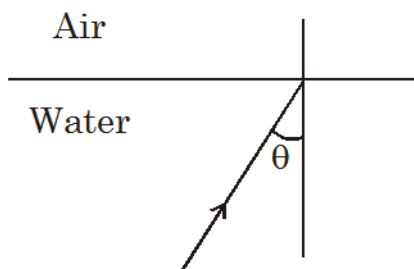
18. A point charge $-q$ is placed inside cavity of a metallic sphere of radius R . Radius of the cavity is r . Potential at point P will be -



- (A) $\frac{-q}{4\pi\epsilon_0 y}$
 (B) $\frac{-q}{4\pi\epsilon_0 y} + \frac{q}{4\pi\epsilon_0 x}$
 (C) $\frac{-q}{4\pi\epsilon_0 x}$
 (D) $\frac{q}{4\pi\epsilon_0 y} - \frac{q}{4\pi\epsilon_0 x}$
19. A particle is projected from bottom of a blind narrow tunnel of depth $\frac{R}{4}$ made along a radius inside the earth. If minimum velocity needed to escape out of the gravitational field of the earth is $\frac{1}{4}\sqrt{\frac{xGM}{R}}$. Find x. M and R are mass and radius of the earth respectively.

- (A) 39
 (B) 78
 (C) 86
 (D) 0

20. A ray of light incidents at an angle of incidence θ in water at water-air interface. Graph between angle of deviation (δ) and angle of incidence (θ) will be-



SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

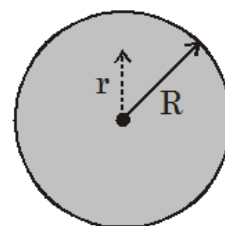
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

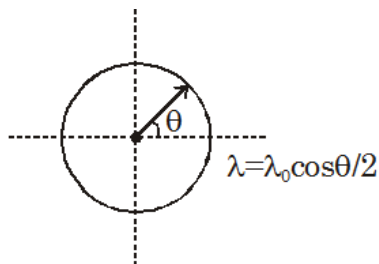
Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

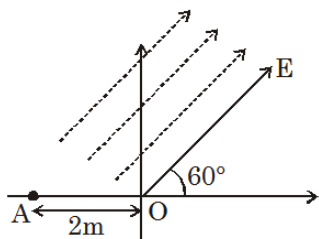
1. A body is thrown from surface of earth with velocity $\left(\sqrt{\frac{g R_e}{2}}\right)$ [R_e radius of earth] at some angle from vertical. Maximum height reach by the body is $\frac{R_e}{4}$. If angle of projection is $\cos^{-1} \left[\frac{\sqrt{x}}{4} \right]$ find the value of x .
2. A uniform solid sphere having mass density is ρ . The ratio of gravitational pressure at a distance $r = R/2$ to the centre of sphere is $\frac{3x}{8}$, the value of x will be



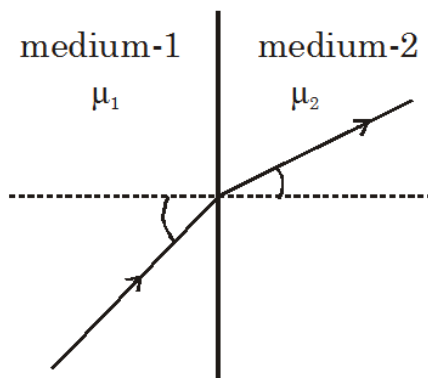
3. Linear charge density of ring of radius 'R' given as $\lambda = \lambda_0 \cos \frac{\theta}{2}$. The potential at centre of ring will be



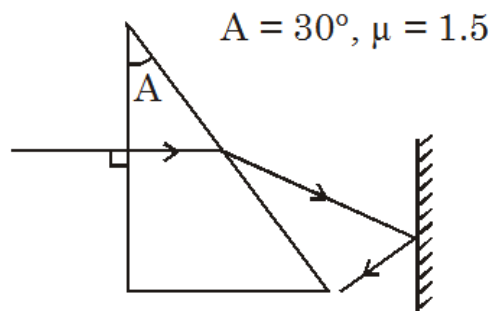
4. Uniform $\vec{E} = 100\text{V/m}$ is directed as shown in figure. The potential difference between $V_A - V_O$ is



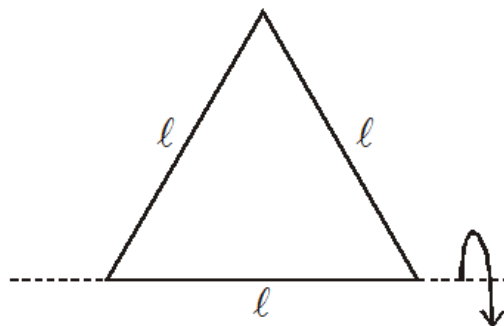
5. A ray incident on medium boundary as shown in figure. Unit vector in direction of incident ray is given as $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ & unit vector of ray in medium (2) is given by $\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}$. The ratio of $\frac{\left(\frac{\mu_1}{\mu_2}\right)^2 + \frac{1}{2}}{\left(\frac{\mu_1}{\mu_2}\right)^2 - \frac{1}{2}}$ is



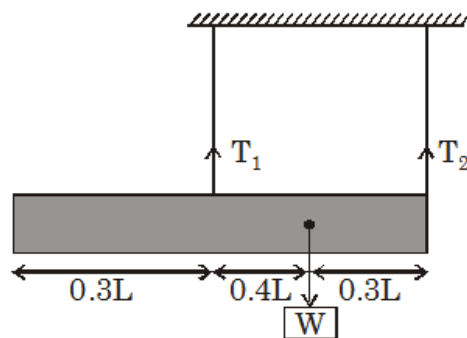
6. A ray incident normal to the surface of prism as shown in figure. Angle of deviation of ray from the mirror w.r.t incident ray is $210^\circ - \sin^{-1} \frac{x}{4}$, find the value of x.



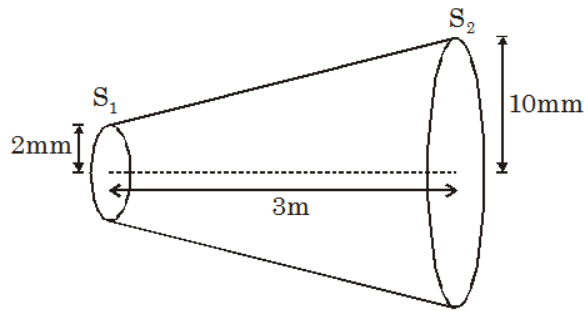
7. Three identical rod each of mass m & length ℓ form a equilateral triangle. Moment of inertia about given axis is $\frac{m\ell^2}{x}$, value of x will be :-



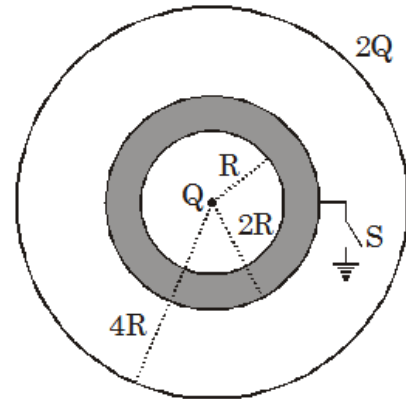
8. In given diagram a uniform plank weight is 500 N. Find out weight W if T_1 & T_2 are equal.



9. Figure shows segment of a current carrying wire made of copper in steady state. If ratio of magnitudes of electric field at cross-section S_1 and S_2 is $5n$. Find value of n .



10. A thick conducting shell of inner radius R and outer radius $2R$ has charge Q at its center. It is covered by a thin conducting shell of radius $4R$ which has charge $2Q$. If charge flow into the earth when switch is closed is nQ , find n .



PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

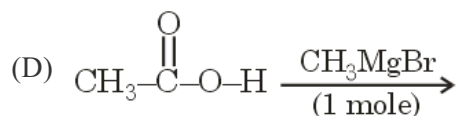
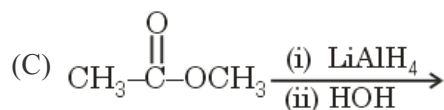
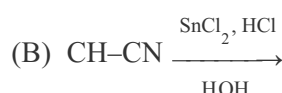
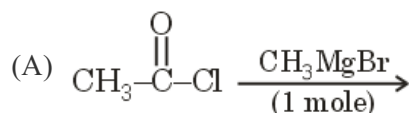
Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

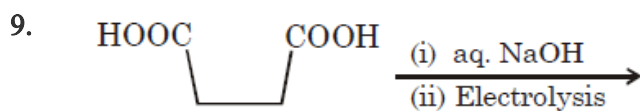
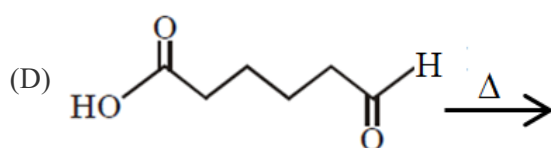
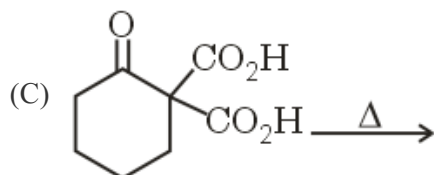
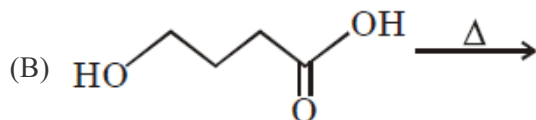
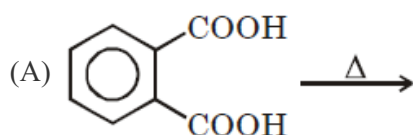
Negative Marks : -1 If wrong option is selected.

- The correct order of second I.E. of C, N, O and F are in the order
(A) $F > O > N > C$
(B) $C > N > O > F$
(C) $O > N > F > C$
(D) $O > F > N > C$
- Calamine malachite, magnetite and cryolite, respectively are:
(A) $ZnSO_4$, $CuCO_3$, Fe_2O_3 , AlF_3
(B) $ZnSO_4$, $Cu(OH)_2$, Fe_3O_4 , Na_3AlF_6
(C) $ZnCO_3$, $CuCO_3 \cdot Cu(OH)_2$, Fe_3O_4 , Na_3AlF_6
(D) $ZnCO_3$, $CuCO_3$, Fe_2O_3 , Na_3AlF_6
- Select the incorrect IUPAC name from the following
(A) $H[AuCl_4]$: Hydrogen tetrachloridoaurate(III)
(B) $[PtCl_6]^{2-}$: Hexachloridoplatinate(IV) ion
(C) $[Cu(NH_3)_4]SO_4$: Tetraamminecopper(II) sulphate
(D) $[Cr(en)_3]Cl_3$:
Tris (ethylenediamine)chromium(III) chloride

- Spin only magnetic moment in BM of $[Fe(CO)_4(C_2O_4)]^+$ is :
(A) 5.92 (B) 0
(C) 2.82 (D) 1.73
- Which of the following complex is optically active :
(A) $trans-[Co(NH_3)_4Cl_2]^+$
(B) $[Cr(H_2O)_6]^{3+}$
(C) $cis-[Co(NH_3)_2(en)_2]^{3+}$
(D) $trans-[Co(NH_3)_2(en)_2]^{3+}$
- Select **INCORRECT** order against property indicated—
(A) $AsCl_3 < SbCl_3 < BiCl_3$ (relative stability)
(B) $ZnO > MgO > CaO > BaO$ (Covalent character)
(C) $NaF < MgF_2 < AlF_3$ (M.P.) Melting point
(D) $BeF_2 > MgF_2 > CaF_2 > SrF_2 > BaF_2$ (water solubility)
- Which of the following gives alcohol as product :



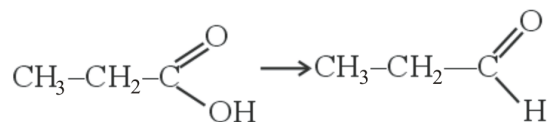
8. Select reaction from the following in which only CO_2 gas is released :



Major product of above reaction :

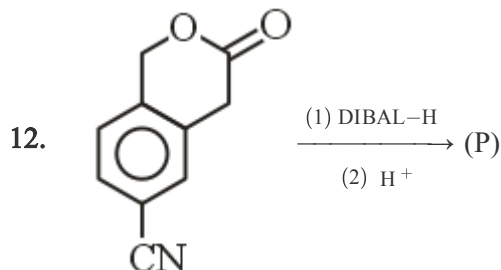
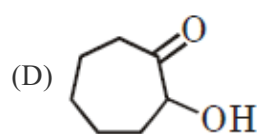
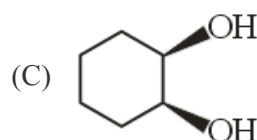
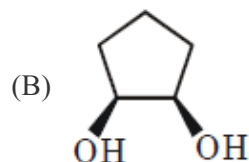
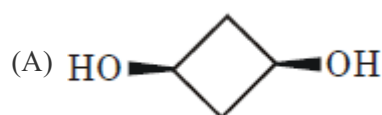
- (A)  (B) $\text{CH}_3\text{-CH}_3$
(C) $\text{CH}_2\text{=CH}_2$ (D) $\text{HC}\equiv\text{CH}$

10. Following conversion can not be carried out by using sequence

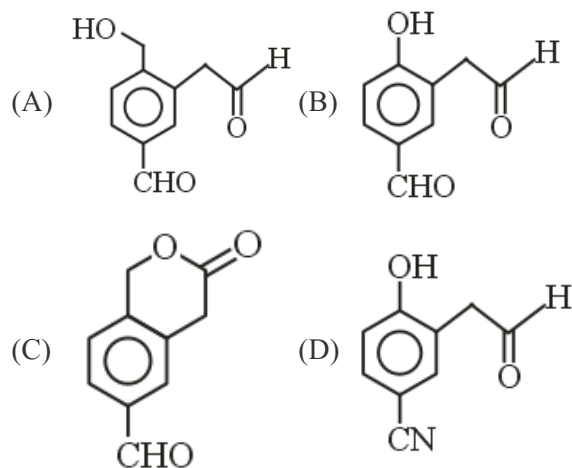


- (A) (i) LiAlH_4 , PBr_3 , DMSO / NaHCO_3 / Δ
(B) (i) SOCl_2 (ii) CH_3MgX (excess)
(C) (i) Ca(OH)_2 (ii) $(\text{HCOO})_2\text{Ca}$ / Dry distillation
(D) (i) LiAlH_4 (ii) $\text{Cu}/300^\circ\text{C}$

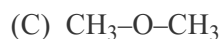
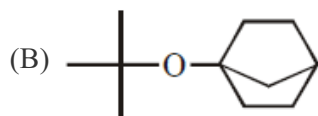
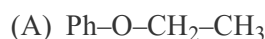
11. Which of the following will not undergo periodic cleavage with HIO_4 , Δ ?



Product P is?



13. Which of the following ether can not be synthesised by Williamson ether synthesis in good yield :



14. At 300 K, the vapour pressure of an ideal solution containing 3 mole of A and 2 mole of B is 600 torr. At the same temperature, if 1.5 mole of A & 0.5 mole of C (non-volatile) are added to this solution the vapour pressure of solution increases by 30 torr. What is the value of P_B° ?

(A) 940 torr

(B) 405 torr

(C) 90 torr

(D) 100 torr

15. What is the correct sequence of osmotic pressures of 0.01 M aq. solutions of :

(i) $\text{Al}_2(\text{SO}_4)_3$ (π_1)

(ii) Na_3PO_4 (π_2)

(iii) BaCl_2 (π_3)

(iv) Glucose (π_4)

[Assume 100% ionisation of electrolytes]

(A) $\pi_4 > \pi_2 > \pi_3 > \pi_1$

(B) $\pi_3 > \pi_4 > \pi_2 > \pi_1$

(C) $\pi_3 > \pi_4 > \pi_1 > \pi_2$

(D) $\pi_1 > \pi_2 > \pi_3 > \pi_4$

16. Find ratio of mole fractions of gases N_2 & O_2 dissolved in water from air at 25° & 1 atm. Assume air is 20 % by volume of O_2 & 80% by volume of N_2

Given: $K_H(\text{N}_2) = 2 \times 10^4 \text{ atm}$

$K_H(\text{O}_2) = 10^4 \text{ atm}$

(A) 8 : 1 (B) 1 : 8 (C) 2 : 1 (D) 1 : 1

17. For a process $\text{A} + \text{B} \rightarrow \text{P}$, the reaction is second order with respect to A and zero order with respect to B. When 1 mole each of A & B are taken in 1 litre vessel the initial rate is $1 \times 10^{-2} \text{ mol/L-sec}$. The rate of reaction when 50% of the reactant have been converted to product would be -

(A) $1 \times 10^{-2} \text{ mol/L-sec}$

(B) $2.5 \times 10^{-3} \text{ mol/L-sec}$

(C) $5 \times 10^{-2} \text{ mol/L-sec}$

(D) $5 \times 10^{-3} \text{ mol/L-sec}$

18. A vessel contains He gas at 20 atm and CH_4 gas at 'P' atm. When a pin-hole is made in the vessel, the gases coming out initially contains 80% He gas, by mole. The value of 'P' is :-

(A) 1 atm (B) 40 atm

(C) 2.5 atm (D) 10 atm

19. A real gas behaves like an ideal gas at 132°C in low pressure range. Critical temperature for this gas is -

(A) 120 K (B) 39 K (C) 132 K (D) 15 K

20. In a CCP lattice of X & Y, X atoms are present at the corners while Y atoms are at face centers. Then the formula of the compound would be if one of the X-atoms from a corner in each unit cell is replaced by Z-atom :

(A) $\text{X}_7\text{Y}_{24}\text{Z}_2$ (B) $\text{X}_7\text{Y}_{24}\text{Z}$

(C) $\text{X}_{24}\text{Y}_7\text{Z}$ (D) $\text{X}\text{Y}_{24}\text{Z}$

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

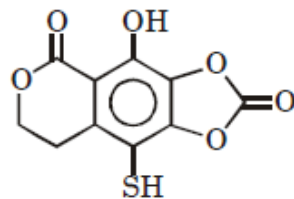
Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

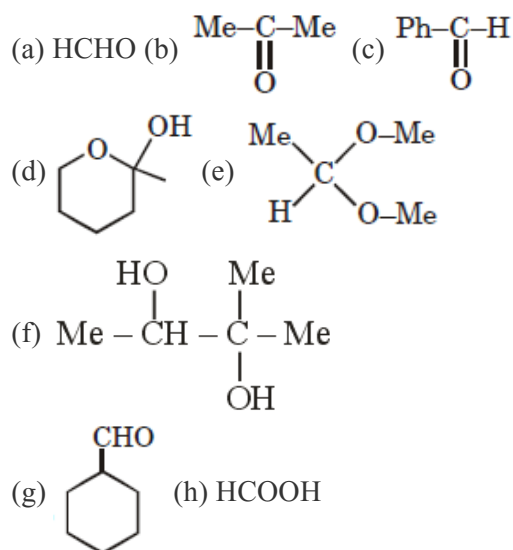
Negative Marks : -1 If wrong option is selected.

- Find out the no. of correct statements :
 (a) Bond length $N_2^+ >$ Bond length N_2
 (b) Bond length $NO^+ <$ Bond length of NO
 (c) Bond length $CN^- <$ Bond length of CN
 (d) Bond length $O_2^- <$ Bond length of O_2^{-2}
 (e) Bond length $O_2 >$ Bond length of O_2^+
 (f) Bond length $B_2 >$ Bond length of B_2^-
- The number of $2c - 2e$ bonds present in a dimer molecule of $Al(CH_3)_3$ is-
- Find the total number of type of isomerism (from given below) shown by the complex $[Fe(NH_3)_4(H_2O)Br](NO_3)_2$
 Ionization isomerism, linkage isomerism, ligand isomerism, Hydrate isomerism, geometrical isomerism
- Total number of geometrical isomers for the complex $[RhCl(CO)(PPh_3)(NH_3)]$ is:

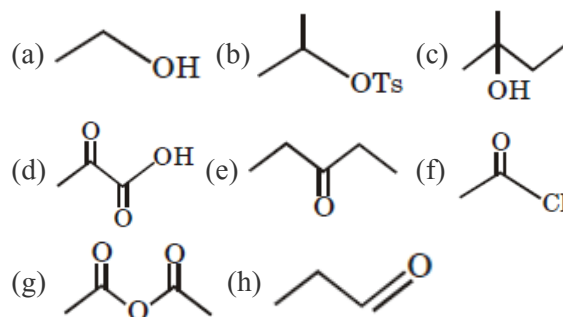
- Calculate total number of moles of Grignard ($MeMgCl$) required for complete reaction with following compound ?



- How many of the following will give positive Fehling test :



- How many of the following will not give iodoform test :



8. For a reaction $A \rightarrow P$
Half life measured for two different values of initial concentrations of A, 5×10^{-3} M and 25×10^{-4} M are 1 and 8 hrs respectively. If initial concentration is adjusted to 1.25×10^{-3} M, what is the half-life? (in hrs)
9. In the following reaction : $xA \longrightarrow yB$, following equation is given :

$$\log \left[-\frac{d[A]}{dt} \right] = \log \left[\frac{d[B]}{dt} \right] + 0.3,$$
 Find ratio $\left(\frac{y}{x} \right)$: [Given $\log 2 = 0.3$]
10. A metal 'A' exists as BCC structure having uncovered length between the adjacent atoms along the edge in an unit cell equal to $1.2 \text{ \AA}$. If molar mass of the metal is 307.2 gm/mole then calculate density of the crystal in gm/ml. [Use : $N_A = 6 \times 10^{23}$ and $\sqrt{3} = 1.7$]

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- $$\lim_{n \rightarrow \infty} \frac{\sum_{r=0}^n \tan^{-1}(1+r+r^2)}{n}$$
 is equal to-
 (A) 1 (B) 2 (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$
- The complete set of values the parameter 'a' so that the point $P\left(a, \frac{1}{1+a^2}\right)$ doesn't lie outside the triangle formed by the lines $15y = x + 1$, $78y = 118 - 23x$ and $y + 2 = 0$ is -
 (A) (0, 5) (B) [2, 5]
 (C) (1, 5) (D) [0, 2]
- Which of the following is true-
 (A) If number of elements in set A are finite, such that $f : A \rightarrow A$ is an one-one function, then f is necessarily onto.
 (B) If a function is continuous in its domain and changes its sign for any 2 values of x then odd number of roots lie between the given values of x
 (C) If $f : A \rightarrow A$ is one-one then it must be onto
 (D) It is possible that a curve contains a local maxima and a global minima at the same point.

- $$\int \frac{2(x^3 - 1)}{x(2x^3 + 1)} dx$$
 is equal to- (where C is the constant of integration)
 (A) $\ln\left(2x + \frac{1}{x^2}\right) + C$
 (B) $\ln(2x^3 + 1) + C$
 (C) $\tan^{-1} \frac{1}{x^{3/2} + 1} + \ln\left(\frac{x-1}{x+1}\right) + C$
 (D) $\ln\left(x + \sqrt{\frac{x-1}{x+1}}\right) + C$
- If the function $g(x) = \begin{cases} ae^x, & x \leq 0 \\ b \cos x + x, & x > 0 \end{cases}$ is differentiable, then the value of $a^2 + b^2$ is -
 (A) 5 (B) 2
 (C) 1 (D) 13
- The number of real solutions of the equation $\tan(e^x) = e^x + e^{-x}$, $x > 0$, is
 (A) 0 (B) 1
 (C) 2 (D) none of these
- If the normal to the curve $y = f(x)$ at the point (4, 6) makes an angle $\frac{2\pi}{3}$ with the positive x axis in anticlockwise direction, then $f'(4)$ is
 (A) $\sqrt{3}$ (B) $-\sqrt{3}$
 (C) $\frac{1}{\sqrt{3}}$ (D) $-\frac{1}{\sqrt{3}}$
- $$\lim_{x \rightarrow 0} \frac{5 \sin x + x \cos x}{2 \tan x - x^2}$$
 is
 (A) $\frac{5}{2}$ (B) 3
 (C) 0 (D) does not exist

9. If linear functions $f(x)$ and $g(x)$ satisfy $\int [(1-2x)\cos x + (3+2x)\sin x] dx = f(x)\sin x + g(x)\cos x + C$ (where C is constant of integration), then
- (A) $f(x) = 3 - 2x$
 (B) $g(x) = 5 - 2x$
 (C) $f(x) = 3 + 2x$
 (D) $g(x) = -5 + 2x$
10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = ax^3 + bx^2 + cx + d$, has no extreme value, then which of the following is always correct ?
- (A) $3a + 2b + c \geq 0$
 (B) $c \leq 0$
 (C) $(3a + 2b + c)c \geq 0$
 (D) $3a + 2b + 2c \geq 0$
11. If $f(x) = 3 - x^2$, $1 \leq x \leq 4$, then the domain of $\log_e(f(2x))$ is
- (A) $\left(-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$ (B) $\left[\frac{1}{2}, 2\right]$
 (C) $\left(0, \frac{\sqrt{3}}{2}\right)$ (D) $\left[\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
12. If two adjacent sides of a parallelogram are given by $2x^2 + 2y^2 - 5xy = 0$ and $5x + 2y - 1 = 0$ is one of its diagonal, then area of parallelogram is $\frac{1}{k}$ sq. units, where k is
- (A) 16 (B) 72 (C) 18 (D) 36
13. Number of positive integral values of 'k' for which the equation $k = |x + |2x - 1|| - |x - |2x - 1||$ has exactly three real solutions, is
- (A) 0 (B) 2 (C) 3 (D) 5
14. $\int (6x^2 + 5x + 4)(x^2 + x + 1)^6 \cdot x^{27} dx$ equals (where C is integration constant)
- (A) $\frac{x^{28}(1+x+x^2)^7}{7} + C$
 (B) $\frac{x^{24}(1+x+x^2)^7}{7} + C$
 (C) $\frac{x^{24}(1+x+x^2)^6}{7} + C$
 (D) $\frac{x^{28}(1+x+x^2)^8}{7} + C$
15. The value of $\lim_{x \rightarrow \infty} x^{\frac{1}{3}} \left((x+1)^{\frac{2}{3}} - (x-1)^{\frac{2}{3}} \right)$ is
- (A) $\frac{4}{3}$ (B) $-\frac{1}{3}$ (C) $\frac{1}{3}$ (D) $-\frac{2}{3}$
16. In the interval $(0, 1)$ maximum value of the function $f(x) = |x \ln x|$ is
- (A) 0 (B) 1 (C) e (D) e^{-1}
17. If the equation $\sin^{-1}(x-1) + \cos^{-1}(x-3) + \tan^{-1}\left(\frac{x}{-x^2+2}\right) = m$ holds, then the value of 'm' is
- (A) $-\frac{3\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $-\frac{5\pi}{4}$ (D) $\frac{5\pi}{4}$
18. The expression $x^3 - 3x^2 - 9x + c$ can be written in the form $(x-a)^2(x-b)$ if the values of c is
- (A) -5 or 27 (B) 5 or -27
 (C) 5 or 27 (D) -5 or -27
19. A real valued function $y = f(x)$ satisfies the relation $f\left(x - \frac{4}{9}\right) + 2x \leq \frac{9}{4}x^2 + \frac{8}{9} \leq f\left(x + \frac{4}{9}\right) - 2x$. The value of $f''(2)$ is
- (A) 4 (B) $\frac{9}{2}$ (C) $\frac{15}{2}$ (D) $\frac{27}{2}$

20. Number of points where the function $f(x) = \max(\sqrt{2x - x^2}, 2 - x)$ is non differentiable, is
- (A) 0 (B) 1
(C) 2 (D) 3

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- If m is the least value of $f(x) = \sqrt{x^2 - 2x + 2} + \sqrt{x^2 - 4x + 29}$, then $[m]$ is equal to (where $[.]$ denotes greatest integer function)
- If 'a' and 'b' are the greatest and least values of $f(x) = 18x^{4/3} - 9x^{1/3} + 1$ in the interval $[-1, 8]$ then the value of $(a + 112b)$ is
- If $f(x) = 10x - 2x^2$, then $\lim_{h \rightarrow 0} \frac{f(3 \cosh h + 4 \sinh h - 2) - 8}{f(3e^h - 5 \sec h + 4) - 12}$ is equal to
- If $f(x) = \begin{cases} \frac{e^{ax} + \ell n(1+4x)+b}{x^3} & x \neq 0 \\ L & x = 0 \end{cases}$ is continuous at $x = 0$, then $12abL$ is equal to

- If $y = \left(\ln(x + \sqrt{x^2 + 1}) \right)^2$ and $(x^2 + 1) \frac{d^3y}{dx^3} + (px + q) \frac{d^2y}{dx^2} + r \frac{dy}{dx} = sy$, then value of $(p + q + r + s)$ is
- $f(x) = \begin{cases} (x+1)^2 & -3 < x \leq 0 \\ 2x^{1/3} + a & 0 < x \leq 1 \\ 2-x & 1 < x < 2 \end{cases}$ has maxima at $x = 0, 1$. If number of integral values of 'a' is L and total number of extrema of $f(x)$ is M then $(L + M)$ is
- If $f(x), g(x), h(x)$ be such that $f'(x) = g(x), g'(x) = h(x), h'(x) = f(x), f(0) = 4, g(0) = 5, h(0) = 0$ then the value of $(f^3(x) + g^3(x) + h^3(x) - 3f(x).g(x).h(x))$ at $x = 2012$ is
- Let $y = f(x)$ be a curve passing through $P(0,1)$ & $Q\left(\frac{1}{2}, \frac{17}{16}\right)$ and having a property that slope of the tangent at any point is proportional to the cube of the abscissa at the point. If the length of normal at Q be ℓ then $\frac{32\ell}{\sqrt{5}}$ is equal to
- Number of integers in the range of function $f(x) = \frac{100}{3(x^6 + 12x^4 + 5x^2 + 1)}$ is
- If $\int \frac{\sqrt{e^{7x}}}{\sqrt[4]{e^{2x} + e^{-2x}}} dx = \frac{1}{k} (e^{\alpha x} + 1)^{\beta/\gamma} + C$ (where C is constant of integration & k, α, β, γ are positive integers and β, γ are relatively prime) then $\left(\frac{k}{\beta} + \frac{\gamma}{\alpha}\right)$ is

Space for Rough Work

